

Collecting Like Terms

KS3 Algebra · Year 7 · Simplifying expressions

Name: _____ Class: _____ Date: _____

HOW TO SIMPLIFY

Collect like terms by **adding or subtracting the coefficients** (the numbers in front). Keep the letter part the same. Like terms have the **same letter and the same power** — so $5m^2$ and $3m^2$ are like terms, but $2m$ is *not*.

WORKED EXAMPLES

Basic

$$5x + 2x = 7x$$

The hidden 1

$$a \text{ means } 1a, \text{ so } 4a + a = 5a$$

$$4a + 5a + a = 10a$$

$$4p + 3q + p = 5p + 3q$$

NEGATIVE TERMS

The sign belongs to the term.

$$-3x - 5x \text{ means } -3x + (-5x) = -8x$$

You are adding two negative amounts, so the answer gets *more* negative.

$$3x - -5x \text{ means } 3x + 5x = 8x$$

Subtracting a negative is the same as adding.

QUESTION 1 — SPOT THE LIKE TERMS

Circle each group of like terms using a different shape. Draw a **circle** around the squared terms (m^2 , t^2 ...) and a **square / box** around the single-letter terms (m , t ...). Some expressions have more than one group.

a) $5m^2 + 2m + 3m^2$

b) $4a + 5a + a$

c) $4p + 3q + p$

d) $7x + 2y + x + 3y$

e) $6t^2 + 2t + t^2 + 5t$

f) $8c + 3d - 2c + d$

Collecting Like Terms — Practice

Simplify, then check your signs and any hidden 1s

QUESTION 2 — SIMPLIFY

a) $4a + 5a + a$

b) $4p + 3q + p$

c) $-3x - 5x$

d) $3x - -5x$

e) $8y - 3y + y$

f) $5m^2 + 2m + 3m^2$

CHALLENGE — WORD PROBLEMS

a) A box contains x red counters, $2x$ blue counters and 5 green counters. Write a simplified expression for the total number of counters.

b) A gardener plants x tulips, $2x$ daffodils and $3x$ rose bushes. She then plants **4 more rose bushes only**. Write a simplified expression for the total number of plants.

QUESTION 3 — SPOT THE MISTAKE

For each one, write the **correct answer** and a short **reason** explaining the mistake.

Sam writes: $3x + 2 = 5x$

Correct answer: _____

Reason: _____

Priya writes: $3x - -5x = -2x$

Correct answer: _____

Reason: _____

Leo writes: $5m^2 + 3m^2 = 8m^4$

Correct answer: _____

Reason: _____